

The Infrastructure You Need for Platform-Centric I&O

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Initiatives: Functional Design

The wide range of new infrastructure options and traditional technologies makes it difficult to decide what is necessary, what is optional, and what is a distraction. This research provides guidance to heads of I&O on where different infrastructure technologies best fit into infrastructure platform plans.

More on This Topic

This is part of an in-depth collection of research. See the collection:

- [Blueprint for Platform-Centric I&O](#)

Overview

Key Findings

- Many infrastructure technologies that vendors label as platforms and promote to heads of infrastructure and IT operations (I&O) are not vital to infrastructure platform success, increasing the risk that poor choices will waste time and resources that are in short supply.
- The tendency for digital product teams to prefer the technology they want can easily lead to situations where little to no platform reuse occurs, limiting the value that heads of I&O derive from infrastructure platform investments.
- Heads of I&O often lack clarity on what existing infrastructure is platform-ready and which requires attention to effectively support digital products.

Recommendations

Heads of I&O whose strategy is to maximize the value of infrastructure platforms for digital products must:

- Adopt an “anchoring” platform strategy by working with heads of EA to encourage each infrastructure platform’s design to center around one technology solution that can provide most of the platform’s required capabilities.
- Limit wasteful platform purchasing by requiring the reuse of complementary capabilities from other platforms already deployed within the organization that meet minimum requirements before making supplemental investments.
- Make existing infrastructure platform-friendly by prioritizing interoperable, automatable, programmable, and intelligent infrastructure investments for existing I&O assets and services.

Strategic Planning Assumption

Through 2028, organizations that use an anchoring strategy for infrastructure platforms will spend 30% less on those technologies compared to those that do not.

Introduction

This note is part of the [Blueprint for Platform-Centric I&O](#), a collection of research that helps heads of I&O set up their organizations to manage the complexity of infrastructure platforms underpinning digital products. This guidance is intended for heads of I&O functions that are successfully optimizing the experience of end users and achieving their negotiated cost and quality goals. Clients that are not yet delivering this level of performance should also consult the [Strategic Roadmap for Platform-Centric I&O](#) for insight into how to address the performance gaps that will limit the value of infrastructure platform adoption.

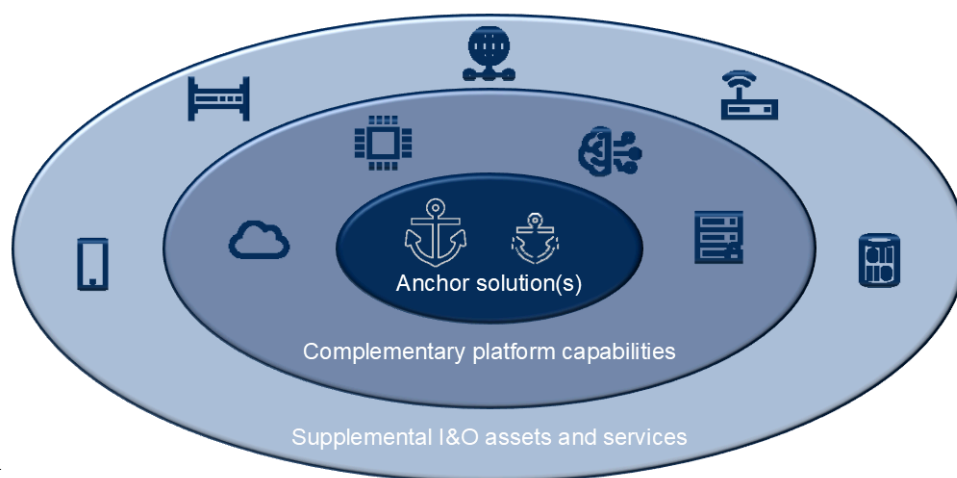
Infrastructure platforms are more than just a bunch of technology assets and services available in one place. By making their capabilities easily consumable via self-service, infrastructure platforms make it easy for each digital product to get what it needs, when it needs it. This curated experience introduces economies of scale, reduces cognitive load, and enables other technical teams to work on higher-level activities with substantial autonomy.

Building an infrastructure platform that meets this expectation is quite difficult, so heads of I&O will typically seek out providers to either provide a needed platform in its entirety or provide essential capabilities for this work. However, lots of technologies are promoted as “platforms” that do not meet this standard. This puts the heads of I&O at high risk of making infrastructure platform purchasing decisions that waste money and fail to provide the resilience and agility required by digital products.

To maximize the value that infrastructure platforms provide to the enterprise, heads of I&O must work with the heads of EA to adopt a sound infrastructure architecture strategy for infrastructure platforms. This entails basing investments for each needed infrastructure platform around a primary domain technology solution, defining clear architectural standards for the use of primary and complementary platform capabilities, and prioritizing investment in interoperable, automatable, programmable, and intelligent infrastructure for the roadmaps of existing I&O assets and services (see Figure 1).

Figure 1: Anatomy of an Infrastructure Platform

Figure 1: Anatomy of an Infrastructure Platform



Source: Gartner
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Analysis

Adopt an “Anchoring” Platform Strategy

Use a small number of solutions (ideally, just one) to “anchor” each infrastructure platform to be delivered, with the anchor providing at least 80% of the platform’s intended functionality. This reduces the complexity of integrating multiple services and disparate capabilities into a single, easily consumable offering. Consult Table 1 for a list of common infrastructure platforms and their potential anchors. This list does not include platforms for operations management (such as observability and ITSM tools), which are discussed in [The Essential Delivery Models and IT Tools for Platform-Centric I&O](#).

Table 1: Common Infrastructure Platforms and Potential Anchors

(Enlarged table in Appendix)

Infrastructure platform	Gartner research on potential anchor options
Compute	Market Guide for Server Virtualization Platforms Market Guide for Edge Computing Platforms Market Guide for Full-Stack Hyperconverged Infrastructure Software
Network	Magic Quadrant for SASE Platforms Magic Quadrant for Enterprise Wired and Wireless LAN Infrastructure Market Guide for Network Automation Platforms Market Guide for Full-Stack Hyperconverged Infrastructure Software
Storage	Magic Quadrant for Enterprise Storage Platforms Market Guide for Full-Stack Hyperconverged Infrastructure Software
Communications	Magic Quadrant for Communications Platform as a Service
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Public cloud	Magic Quadrant for Strategic Cloud Platform Services Market Guide for Industry Cloud Platforms
Hybrid/private cloud	Magic Quadrant for Distributed Hybrid Infrastructure Market Guide for Platform-Native Consumption Services

Source: Gartner (December 2025)

When adopting an anchor infrastructure platform strategy, heads of I&O must actively manage the risk of vendor lock-in. This is best handled by taking the following actions:

- Identify systematic barriers to changing vendors, such as contractual obligations, employee skill set limitations, capability differences, ecosystem dependencies, compatibility with management tooling and data portability.

- Document and evaluate the risks posed by dependence on the vendor, including threats to business continuity, governance challenges and future incompatibility with business goals.
- Create a vendor risk mitigation strategy that outlines options for reducing the dependency on the vendor, and the trade-offs the organization is willing to make to modify its relationship with the vendor when warranted.
- Include guidance on managing vendor lock-in in the platform strategy, specifying when a platform must be built to accommodate changes to the “anchor” offering and guidance to the platform team on how to reduce lock-in risk to acceptable levels.

For an example of how this approach can be applied, see [Quick Answer: How Should Executive Leaders Respond to Cloud Vendor Lock-In Concerns?](#)

Require Reuse of Complementary Platform Capabilities

To limit duplicate spending on functionality not provided by an anchor offering, heads of I&O must insist on using other platform technology capabilities as the primary means of complementing anchor investments. Infrastructure platforms are not the only types of platforms that can be valuable to I&O. Work with digital product and IT function leaders to provide visibility into the total set of platform capabilities available to platform engineers across the organization. Consult Table 2 for a list of common platforms that can be useful to I&O for enhancing infrastructure platforms. This list does not include platforms for operations management (such as observability and ITSM tools), which are discussed in [The Essential Delivery Models and IT Tools for Platform-Centric I&O](#).

Table 2: Common Platforms With Useful Capabilities for Infrastructure Platform Enhancement

(Enlarged table in Appendix)

Platform	Relevant Gartner resources
Security	Innovation Insight for Security Platforms Magic Quadrant for Backup and Data Protection Platforms Magic Quadrant for CPS Protection Platforms Market Guide for Cloud-Native Application Protection Platforms
Application	Magic Quadrant for Cloud-Native Application Platforms Magic Quadrant for Enterprise Low-Code Application Platforms Magic Quadrant for Cloud-Native Application Platforms Magic Quadrant for Integration Platform as a Service
Digital experience	Magic Quadrant for Digital Experience Platforms Magic Quadrant for Digital Employee Experience Management Tools Magic Quadrant for Endpoint Protection Platforms
Data and analytics	Magic Quadrant for Analytics and Business Intelligence Platforms Magic Quadrant for Data and Analytics Governance Platforms Market Guide for Conversation Analytics Platforms
AI	Magic Quadrant for Data Science and Machine Learning Platforms Magic Quadrant for Conversational AI Platforms Market Guide for AI Application Development Platforms
Note: Some recommended content may not be available as part of your current Gartner subscription.	

Work closely with the heads of EA to make reference architectures (RA) available so that infrastructure platform teams can reuse functionality already deployed in other platforms. Be an advocate for this work:

- Serve on RA committees.
- Encourage infrastructure platform teams to provide feedback on RA quality and ease of use.
- Sponsor pilots for infrastructure platform teams to increase decision making, and autonomy in using RA components.

Consult [Build Resilient, Future-Ready Reference Architectures](#) for more on this topic and how RAs can improve reuse and reduce the cost of infrastructure platform delivery.

Make Existing Infrastructure Platform-Friendly

Platform-centric I&O still requires the use of infrastructure assets and services, yet these systems will need changes to enable the success of these efforts. Infrastructure and services must exhibit these characteristics to support digital product and infrastructure platform efforts:

- **Interoperability:** The ability for a system from one manufacturer to work with one from another source. Infrastructure that cannot seamlessly coordinate with other systems cannot be scaled efficiently and is of little use to an infrastructure platform as a result.
- **Automatability:** The degree to which the system can be performed or controlled by other technology with minimal or no human intervention. Infrastructure that is not easily automated will exhibit inconsistencies that undermine resilience and trust, limiting the number of activities that an infrastructure platform can safely undertake.
- **Programmability:** The extent to how well tools and methods from software development can be used to adjust system performance through programmable interfaces. Infrastructure that is not programmable makes it difficult to change aspects of system performance that are necessary to meet platform needs, resulting in a loss of value or increased costs and risks to platform outcomes.
- **Intelligence:** The degree to how well a system can optimize its own performance in response to changes in demand is what intelligence refers to in this context. Infrastructure that is not intelligent slows down the response of infrastructure platforms to digital product needs, which ultimately impacts the ability of that product to support enterprise customers.

Heads of I&O must direct their teams to account for these factors in infrastructure life cycle planning. Not every infrastructure component must fully embody all of the characteristics outlined above. It is essential, though, that technology asset and infrastructure service owners assess systems they are responsible for in order to understand potential impacts on infrastructure platforms and plan to remediate critical gaps identified. For more on specific technologies and recommended reference architectures to guide selection, consult Gartner's [Reference Architecture Resource Center](#). For guidance on strengthening these practices, see [How to Reduce and Manage Infrastructure Technical Debt](#).

By adopting an anchoring strategy for infrastructure platforms, and requiring the reuse of complementary platform capabilities and making existing infrastructure platform-friendly, heads of I&O can minimize the investments required to build and operate effective infrastructure platforms that support digital transformation.

Evidence

The positions taken in this note were informed by analysis of responses to Gartner's IT Score for Infrastructure and IT Operations. The IT Score for Infrastructure and IT Operations offers the best of Gartner expertise and peer-based research. Maturity is measured on a scale ranging from 1 (low) to 5 (high). It measures how advanced an organization's development is in a functional activity relative to Gartner's expert assessment of practices associated with that activity. The maturity of an activity is measured by several subactivities weighted 1 through 5, depending on how progressive they are. Importance is measured on a scale of 1 (low) to 5 (high). Importance measures how important each functional activity is to the overall effectiveness of the function in meeting its business objectives. 341 organizations completed the IT Score for Infrastructure & IT Operations assessment online from September 2024 through June 2025.

Recommended by the Author

Some documents may not be available as part of your current Gartner subscription.

[What Good Modern Infrastructure Looks Like](#)

[Hype Cycle for Infrastructure Platforms, 2025](#)

[Cool Vendors in Platform Engineering for Abstracting Infrastructure Complexity](#)

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